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LIGHT-EMITTING DEVICE AND DISPLAY DEVICE INCLUDING THE SAME

Abstract

A light-emitting device is provided. The device includes: a semiconductor light-emitting structure including a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer which are sequentially provided in a first direction; and an electrode layer covering the second conductivity type semiconductor layer and spaced apart from the active layer in the first direction with the second conductivity type semiconductor layer therebetween. The semiconductor light-emitting structure is configured to emit light having a wavelength (λ) within a range from about 490 nm to about 700 nm. A minimum distance from the electrode layer to the active layer in the first direction is greater than about 0.05λ and less than about 0.24λ .

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0032842, filed on Mar. 7, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] The present disclosure relates to a light-emitting device and a display device including the same, and more particularly, to a light-emitting device including an active layer and a display device including the light-emitting device.

[0003] Light-emitting diodes (LEDs) light sources which convert electrical energy into optical energy and are widely used as a light source for various display devices, such as a lighting system, a television, a mobile phone, a personal computer, a laptop computer, a personal digital assistant (PDA), a digital camera, a camcorder, a viewfinder, a micro display, a three-dimensional (3D) display, a virtual reality or augmented reality display, etc. Recently, micro- or nano-scale subminiature LEDs using II-VI group or III-V group compound semiconductors have been developed, and accordingly, there has been a growing demand for a light-emitting device having a novel structure to improve light extraction efficiency in such subminiature LEDs.

SUMMARY

[0004] One or more embodiments provide a light-emitting device having a structure capable of improving the light extraction efficiency.

[0005] One or more embodiments also provide a display device including a light-emitting device having a structure capable of improving the light extraction efficiency.

[0006] According to an aspect of an embodiment, a light-emitting device includes: a semiconductor light-emitting structure including a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer which are sequentially provided in a first direction; and [0007] an electrode layer covering the second conductivity type semiconductor layer and spaced apart from the active layer in the first direction with the second conductivity type semiconductor layer therebetween, [0008] wherein the semiconductor light-emitting structure is configured to emit light having a wavelength (λ) within a range from about 490 nm to about 700 nm, and [0009] wherein a minimum distance from the electrode layer to the active layer in the first direction is greater than about 0.05λ and less than about 0.24λ .

[0010] According to another aspect of an embodiment, a light-emitting device includes: a first conductivity type base semiconductor layer; a semiconductor light-emitting structure including a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer which are sequentially provided on a main surface of the first conductivity type base semiconductor layer in a first direction perpendicular to the main surface; a first electrode surrounding a side wall of the first conductivity type base semiconductor layer; a transparent electrode layer covering the second conductivity type semiconductor layer and spaced apart from the active layer in the first direction with the second conductivity type semiconductor layer therebetween; a reflective structure covering a side wall of each of the semiconductor light-emitting structure and the transparent electrode layer; a second electrode in contact with the transparent electrode layer and spaced apart from the second conductivity type semiconductor layer

in the first direction with the transparent electrode layer therebetween; and a microlens in contact with an emission surface including a part of a rear surface opposite to the main surface of the first conductivity type base semiconductor layer and configured to collect and emit light from the semiconductor light-emitting structure. The semiconductor light-emitting structure is configured to emit light having a wavelength (λ) within a range from about 490 nm to about 700 nm. The semiconductor light-emitting structure has a width of 100 μm or less in a second direction parallel with the main surface of the first conductivity type base semiconductor layer. A minimum distance from the active layer to the transparent electrode layer in the first direction is greater than 0.05λ and less than 0.24λ . The active layer includes at least one multi-quantum well layer. A thickness of the at least one multi-quantum well layer in the first direction is less than 300 nm.

[0011] According to another aspect of an embodiment, a display device includes: a circuit board including a driver circuit; and a pixel array on the circuit board and including a plurality of pixels. The pixel array includes: a first conductivity type base semiconductor layer including a main surface and a rear surface which are opposite to each other; a plurality of semiconductor light-emitting structures on the main surface of the first conductivity type base semiconductor layer and spaced apart from each other in a horizontal direction parallel with the main surface, each of the plurality of semiconductor light-emitting structures including a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer which are sequentially provided in a first direction perpendicular to the main surface; a first electrode including a metal layer having a grid shape penetrating the first conductivity type base semiconductor layer in the first direction; a plurality of transparent electrode layers covering the second conductivity type semiconductor layer in each of the plurality of semiconductor light-emitting structures; a reflective structure covering a side wall of each of the plurality of semiconductor light-emitting structures and the plurality of transparent electrode layers; a plurality of second electrodes penetrating the reflective structure in the first direction and in contact with one transparent electrode layer selected from the plurality of transparent electrode layers; and a plurality of microlenses on the rear surface of the first conductivity type base semiconductor layer and overlapping the plurality of semiconductor light-emitting structures in the first direction. At least one semiconductor light-emitting structure selected from the plurality of semiconductor light-emitting structures is configured to emit light having a wavelength (λ) within a range from about 490 nm to about 700 nm. In the at least one semiconductor light-emitting structure, a minimum distance from the active layer to a transparent electrode layer, from among the plurality of transparent electrode layers, in the first direction is greater than about 0.05λ and less than about 0.24λ .

Description

BRIEF DESCRIPTION OF DRAWINGS

[0012] The above and other aspects will be more apparent from the following description of embodiments, taken in conjunction with the accompanying drawings, in which:

[0013] FIG. 1 is a cross-sectional view illustrating a schematic structure of a light-emitting device according to some embodiments;

[0014] FIG. 2A is a schematic perspective view illustrating examples of shapes of a semiconductor light-emitting structure and a transparent electrode layer according to some embodiments, and FIG. 2B is a schematic plan view illustrating the semiconductor light-emitting structure of FIG. 2A according to some embodiments;

[0015] FIG. 3A is a schematic perspective view illustrating examples of shapes of a semiconductor light-emitting structure and a transparent electrode layer which are included in a light-emitting device according to some embodiments, and FIG. 3B is a schematic plan view illustrating a plane

shape of the semiconductor light-emitting structure of FIG. 3A according to some embodiments;

[0016] FIGS. 4 and 5 illustrate a schematic plan views of example plane shapes of a semiconductor light-emitting structures included in light-emitting devices according to some embodiments;

[0017] FIGS. 6, 7, and 8 illustrate cross-sectional views of light-emitting devices according to some embodiments;

[0018] FIG. 9A is a cross-sectional view illustrating a light-emitting device according to some embodiments, and FIG. 9B is an enlarged cross-sectional view illustrating a part denoted by “EX1” in FIG. 9A according to some embodiments;

[0019] FIG. 10 is a view illustrating a relation between a change in light extraction efficiency according to position of an active layer of a semiconductor light-emitting structure included in the light-emitting device of FIGS. 9A and 9B and a position of each of a first multi-quantum well layer and a second multi-quantum well layer which are included in the active layer;

[0020] FIG. 11 is a schematic perspective view illustrating a display device according to some embodiments, FIG. 12 is an enlarged plan view illustrating a part denoted by “EX2” in FIG. 11 according to some embodiments, and FIG. 13 is a cross-sectional view schematically illustrating components of a portion along a cross-section cut by line I-I’ in FIG. 11 and a portion along a cross-section cut by line II-II’ in FIG. 12 according to some embodiments;

[0021] FIGS. 14 and 15 each illustrate a graph showing an simulation result for evaluation of light extraction efficiency of a light-emitting device according to some embodiments;

[0022] FIGS. 16, 17, and 18 each illustrate a graph showing an simulation result for evaluation of light extraction efficiency of a light-emitting device according to some embodiments;

[0023] FIGS. 19A, 19B, 19C, 19D, 19E, 19F and 19G each illustrate a cross-sectional view of a method of manufacturing an display device including a light-emitting device according to a process order, according some embodiments;

[0024] FIG. 20 is a block diagram illustrating an example of an electronic device including a display device or a light-emitting device according some embodiments;

[0025] FIG. 21 is a view illustrating an example of a mobile device as an application example of an electronic device including a display device or a light-emitting device according some embodiments;

[0026] FIG. 22 is a view illustrating an example of a head-up display device for vehicles as an application example of an electronic device including a display device or a light-emitting device according some embodiments;

[0027] FIG. 23 is a view illustrating an example of augmented reality glasses or virtual reality glasses as an application example of an electronic device including a display device or a light-emitting device according some embodiments;

[0028] FIG. 24 is a view illustrating an example of a large-scale signage as an application example of an electronic device including a display device or a light-emitting device according some embodiments; and

[0029] FIG. 25 is a view illustrating an example of a wearable display as an application example of an electronic device including a display device or a light-emitting device according some embodiments.

DETAILED DESCRIPTION

[0030] Hereinafter, embodiments of the inventive concept are described in detail with reference to the accompanying drawings. Embodiments described herein are example embodiments, and thus, the present disclosure is not limited thereto, and may be realized in various other forms. Each embodiment provided in the following description is not excluded from being associated with one or more features of another example or another embodiment also provided herein or not provided herein but consistent with the present disclosure. It will be understood that when an element or layer is referred to as being “on,” “connected to” or “coupled to” another element or layer, it can be directly on, connected or coupled to the other element or layer, or intervening elements or layers

may be present. By contrast, when an element is referred to as being “directly on,” “directly connected to” or “directly coupled to” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. For example, the expression, “at least one of a, b, and c,” should be understood as including only a, only b, only c, both a and b, both a and c, both b and c, or all of a, b, and c. It will be also understood that, even if a certain step or operation of manufacturing an apparatus or structure is described later than another step or operation, the step or operation may be performed later than the other step or operation unless the other step or operation is described as being performed after the step or operation. In the drawings, like reference numerals denote like components, and any redundant description thereon will be omitted.

[0031] FIG. 1 is a cross-sectional view illustrating a schematic structure of a light-emitting device **100** according to some embodiments.

[0032] Referring to FIG. 1, the light-emitting device **100** may include a first conductivity type base semiconductor layer **102** and a semiconductor light-emitting structure **110** arranged on a main surface **102M** of the first conductivity type base semiconductor layer **102**. The semiconductor light-emitting structure **110** may include a first conductivity type semiconductor layer **112**, an active layer **114**, and a second conductivity type semiconductor layer **116**, which are sequentially stacked on the main surface **102M** in a first direction (Z direction in FIG. 1) perpendicular to the main surface **102M** of the first conductivity type base semiconductor layer **102**.

[0033] The semiconductor light-emitting structure **110** may include a micro light-emitting diode (LED). In some embodiments, the semiconductor light-emitting structure **110** may include a micro LED which generates light having a color selected from red, green, and blue. The term “micro LED” as used herein refers to a LED having a width of 100 μm or less in a horizontal direction (e.g., X direction in FIG. 1) perpendicular to the first direction (Z direction in FIG. 1). For example, the width of the semiconductor light-emitting structure **110** in the horizontal direction (e.g., X direction in FIG. 1) may be about 100 μm or less, about 50 μm or less, about 20 μm or less, about 10 μm or less, about 6 μm or less, about 5 μm or less, about 4 μm or less, or about 2 μm or less; however, embodiments are not limited thereto.

[0034] The semiconductor light-emitting structure **110** may be configured to emit light having a wavelength (λ) within a range from about 400 nm to about 700 nm.

[0035] In some embodiments, the semiconductor light-emitting structure **110** may be configured to emit light having a first wavelength (λ_1) within a range from about 580 nm to about 700 nm. The light having the first wavelength (λ_1) may be red light. As used herein, the wavelength region of red light refers to a wavelength region of about 580 nm or more and less than about 700 nm, for example, a wavelength region from about 610 nm to about 650 nm or a wavelength region from about 620 nm to about 640 nm, and there may be at least one emission spectrum peak in the wavelength region of red light.

[0036] In some embodiments, the semiconductor light-emitting structure **110** may be configured to emit light having a second wavelength (λ_2) within a range from about 490 nm to about 580 nm. The light having the second wavelength (λ_2) may be green light. As used herein, the wavelength region of green light refers to a wavelength region of about 490 nm or more and less than about 580 nm, for example, a wavelength region from about 510 nm to about 550 nm or a wavelength region from about 520 nm to about 540 nm, and there may be at least one emission spectrum peak in the wavelength region of green light.

[0037] In some embodiments, the semiconductor light-emitting structure **110** may be configured to emit light having a third wavelength (λ_3) within a range from about 400 nm to about 490 nm. The light having the third wavelength (λ_3) may be blue light. As used herein, the wavelength region of blue light refers to a wavelength region of about 400 nm or more and less than about 490 nm, for

example, a wavelength region from about 440 nm to about 480 nm or a wavelength region from about 450 nm to about 470 nm, and there may be at least one emission spectrum peak in the wavelength region of blue light.

[0038] Each of the first conductivity type base semiconductor layer **102**, the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116** may include an epitaxial nitride semiconductor layer. The first conductivity type base semiconductor layer **102** and the first conductivity type semiconductor layer **112** may include a nitride semiconductor layer doped with the same type of dopant, e.g., n-type dopant, and an average doping concentration of the first conductivity type base semiconductor layer **102** may be higher than an average doping concentration of the first conductivity type semiconductor layer **112**. Each of the first conductivity type semiconductor layer **112** and the second conductivity type semiconductor layer **116** may include a single layer or include multiple layers including a plurality of layers having different doping concentrations of dopant, compositions of constituent components, etc., from each other. The first conductivity type base semiconductor layer **102** may be spaced apart from the active layer **114** in the first direction (Z direction in FIG. 1) with the first conductivity type semiconductor layer **112** arranged therebetween.

[0039] The first conductivity type base semiconductor layer **102** may have a thickness of about 10 nm to about 6,000 nm in the first direction (Z direction in FIG. 1). The first conductivity type semiconductor layer **112** may have a thickness of about 10 nm to about 500 nm in the first direction.

[0040] In the semiconductor light-emitting structure **110**, the first conductivity type semiconductor layer **112** may be integrally connected to the first conductivity type base semiconductor layer **102**. In some embodiments, the first conductivity type base semiconductor layer **102** and the first conductivity type semiconductor layer **112** may include the same material. In some embodiments, the first conductivity type base semiconductor layer **102** may include an n-type gallium nitride (n-GaN). The first conductivity type semiconductor layer **112** may include an n-type superlattice structure layer. For example, the first conductivity type semiconductor layer **112** may include an InGaN/GaN superlattice structure layer. In this case, the first conductivity type semiconductor layer **112** may have a superlattice structure in which an InGaN film and a GaN film are alternately stacked. In the first conductivity type semiconductor layer **112**, the superlattice structure may include a pair structure of an InGaN film and a GaN film in about 10 cycles to about 50 cycles, for example, about 15 cycles to about 20 cycles; however, embodiments are not limited thereto.

[0041] In some embodiments, the first conductivity type semiconductor layer **112** may include a nitride semiconductor layer having a composition of $\text{In}_{0.5x}\text{Al}_{0.5y}\text{Ga}_{1-x-y}\text{N}$ ($0 \leq x < 1$, $0 \leq y < 1$, $0 \leq x+y < 1$). In some embodiments, the first conductivity type semiconductor layer **112** may include an n-type gallium nitride (n-GaN) doped with silicon (Si), germanium (Ge), or carbon (C). In some embodiments, the first conductivity type semiconductor layer **112** may be a semiconductor layer including aluminum indium gallium phosphide (AlInGaP) or aluminum indium gallium arsenide (AlInGaAs).

[0042] In the semiconductor light-emitting structure **110**, the active layer **114** may emit light having certain energy by recombination of electrons and holes. The active layer **114** may have a single-quantum well or multi-quantum well structure in which a quantum barrier layer and a quantum well layer are arranged alternately. In some embodiments, the active layer **114** may have a single- or multi-quantum well structure including a pair structure of one quantum barrier layer and one quantum well layer in 1 to 15 cycles.

[0043] In some embodiments, the active layer **114** may include a quantum barrier layer and a quantum well layer including a compound semiconductor of III-V group elements. For example, the active layer **114** may include a pair structure selected from InGaN/GaN, InGaN/InGaN, InGaN/AlGaN, and InGaN/InAlGaN; however, embodiments are not limited thereto.

[0044] In some embodiments, the quantum well layer and the quantum barrier layer may

respectively include In.sub.xAl.sub.yGa.sub.1-x-yN layers ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$) having different compositions from each other. For example, the quantum well layer may include an undoped In.sub.xGa.sub.1-xN layer ($0 < x < 1$), and the quantum barrier layer may include an undoped GaN layer or a GaN layer doped with silicon (Si).

[0045] In some embodiments, when the quantum barrier layer included in the active layer **114** is an In.sub.xGa.sub.1-xN layer ($0 < x < 1$), the bandgap energy at the active layer **114** may be controlled according to a content ratio of indium (In) in the quantum well layer, thereby controlling an emission wavelength band. When the quantum well layer included in the active layer **114** is an In.sub.xGa.sub.1-xN layer ($0 < x < 1$), a value of x, which represents a content ratio of indium (In) in the quantum well layer, may be within a range from about 0.15 to about 0.35. For example, when the semiconductor light-emitting structure **110** is configured to emit red light, the x value in the In.sub.xGa.sub.1-xN layer ($0 < x < 1$) constituting the quantum well layer included in the active layer **114** may be within a range from about 0.3 to about 0.35, when the semiconductor light-emitting structure **110** is configured to emit green light, the x value in the In.sub.xGa.sub.1-xN layer ($0 < x < 1$) constituting the quantum well layer included in the active layer **114** may be within a range from about 0.25 to about 0.3, and when the semiconductor light-emitting structure **110** is configured to emit blue light, the x value in the In.sub.xGa.sub.1-xN layer ($0 < x < 1$) constituting the quantum well layer included in the active layer **114** may be within a range from about 0.15 to about 0.2; however, embodiments are not limited thereto.

[0046] In the first direction (Z direction in FIG. 1), a thickness **114T** of the active layer **114** may be less than 300 nm. As illustrated in FIG. 1, the active layer **114** may include a surface in contact with the first conductivity type semiconductor layer **112** and a surface in contact with the second conductivity type semiconductor layer **116**, and in the active layer **114**, a minimum distance from the surface in contact with the first conductivity type semiconductor layer **112** to the surface in contact with the second conductivity type semiconductor layer **116** may be less than 300 nm.

[0047] In some embodiments, the active layer **114** may include a multi-quantum well layer including a pair structure of one quantum barrier layer and one quantum well layer in 8 to 12 cycles, wherein the multi-quantum well layer may include a surface in contact with the first conductivity type semiconductor layer **112** and a surface in contact with the second conductivity type semiconductor layer **116**, and in the first direction, the multi-quantum well layer may have a thickness less than 300 nm.

[0048] In some embodiments, the thickness **114T** of the active layer **114** may be less than 300 nm, less than 200 nm, less than 100 nm, less than 50 nm, less than 40 nm, less than 20 nm, less than 10 nm, less than 5 nm, or less than 3 nm. For example, the thickness **114T** of the active layer **114** may be within a range from about 2 nm to about 10 nm; however, embodiments are not limited thereto.

[0049] In some embodiments, the thickness **114T** of the active layer **114** may be determined according to a wavelength (λ) of light emitted from the semiconductor light-emitting structure **110**, and the thickness **114T** of the active layer **114** may be less than 0.4λ . In an embodiment, when the semiconductor light-emitting structure **110** is configured to emit red light having a first wavelength (λ_1) within a range from about 580 nm to about 700 nm, the thickness **114T** of the active layer **114** may be less than $0.4\lambda_1$. In another embodiment, when the semiconductor light-emitting structure **110** is configured to emit green light having a second wavelength (λ_2) within a range from about 490 nm to about 580 nm, the thickness **114T** of the active layer **114** may be less than $0.4\lambda_2$. In another embodiment, when the semiconductor light-emitting structure **110** is configured to emit blue light having a third wavelength (λ_3) within a range from about 400 nm to about 490 nm, the thickness **114T** of the active layer **114** may be less than $0.4\lambda_3$.

[0050] In the semiconductor light-emitting structure **110**, the second conductivity type semiconductor layer **116** may include a nitride semiconductor layer doped with a p-type dopant. In some embodiments, the second conductivity type semiconductor layer **116** may include a nitride semiconductor layer having a composition of In.sub.xAl.sub.yGa.sub.1-x-yN ($0 \leq x < 1$, $0 \leq y < 1$,

$0 \leq x+y < 1$). For example, the second conductivity type semiconductor layer **116** may include a p-type gallium nitride (p-GaN) doped with magnesium (Mg) or zinc (Zn). However, embodiments are not limited thereto. In some embodiments, the second conductivity type semiconductor layer **116** may be a semiconductor layer including aluminum indium gallium phosphide (AlInGaP) or aluminum indium gallium arsenide (AlInGaAs).

[0051] The light-emitting device **100** may include a transparent electrode layer **130** covering the second conductivity type semiconductor layer **116** and a reflective electrode layer **170**. The transparent electrode layer **130** may be arranged to be in contact with the second conductivity type semiconductor layer **116** and spaced apart from the active layer **114** in the first direction (Z direction in FIG. 1) with the second conductivity type semiconductor layer **116** arranged therebetween. The reflective electrode layer **170** may be arranged to be in contact with the transparent electrode layer **130** and spaced apart from the second conductivity type semiconductor layer **116** in the first direction (Z direction in FIG. 1) with the transparent electrode layer **130** arranged therebetween. The transparent electrode layer **130** may be referred to as an electrode layer, and the reflective electrode layer **170** may be referred to as an electrode layer or a second electrode.

[0052] In the first direction, a minimum distance from the reflective electrode layer **170** to the active layer **114** may be determined according to a wavelength (λ) of light emitted from the semiconductor light-emitting structure **110**. The minimum distance from the reflective electrode layer **170** to the active layer **114** in the first direction may be greater than 0.05λ and less than 0.4λ . In some embodiments, the minimum distance from the reflective electrode layer **170** to the active layer **114** in the first direction may be greater than 0.05λ and less than 0.24λ .

[0053] As illustrated in FIG. 1, the second conductivity type semiconductor layer **116** may include a surface in contact with the active layer **114** and a surface in contact with the transparent electrode layer **130**, and in the second conductivity type semiconductor layer **116**, the surface in contact with the active layer **114** may be opposite to the surface in contact with the transparent electrode layer **130** in the first direction. As such, when the second conductivity type semiconductor layer **116** is arranged between the active layer **114** and the transparent electrode layer **130**, and in contact with each of the active layer **114** and the transparent electrode layer **130**, the minimum distance from the reflective electrode layer **170** to the active layer **114** in the first direction may correspond to a thickness **116T** of the second conductivity type semiconductor layer **116** in the first direction, and the thickness **116T** of the second conductivity type semiconductor layer **116** may be greater than 0.05λ and less than 0.4λ . In some embodiments, the thickness **116T** of the second conductivity type semiconductor layer **116** in the first direction may be greater than 0.05λ and less than 0.24λ .

[0054] When the semiconductor light-emitting structure **110** is configured to emit red light having a first wavelength (λ_1) within a range from about 580 nm to about 700 nm, the thickness **116T** of the second conductivity type semiconductor layer **116** in the first direction may be greater than 0.05λ and less than 0.24λ , for example, greater than 0.1λ and less than 0.24λ . For example, the thickness **116T** of the second conductivity type semiconductor layer **116** may be about 29 nm to about 210 nm, about 31 nm to about 180 nm, about 63 nm to about 152 nm, or about 95 nm to about 135 nm; however, embodiments are not limited thereto.

[0055] When the semiconductor light-emitting structure **110** is configured to emit green light having a second wavelength (λ_2) within a range from about 490 nm to about 580 nm, the thickness **116T** of the second conductivity type semiconductor layer **116** in the first direction may be greater than 0.05λ and less than 0.24λ , for example, greater than 0.1λ and less than 0.24λ . For example, the thickness **116T** of the second conductivity type semiconductor layer **116** may be about 24 nm to about 174 nm, about 26 nm to about 159 nm, about 53 nm to about 128 nm, or about 60 nm to about 100 nm; however, embodiments are not limited thereto.

[0056] When the semiconductor light-emitting structure **110** is configured to emit blue light having a third wavelength (λ_3) within a range from about 400 nm to about 490 nm, the thickness **116T** of

the second conductivity type semiconductor layer **116** in the first direction may be greater than 0.05λ and less than 0.4λ , for example, greater than 0.2λ and less than 0.4λ . For example, the thickness **116T** of the second conductivity type semiconductor layer **116** may be about 80 nm to about 196 nm, about 23 nm to about 184 nm, about 92 nm to about 184 nm, or about 125 nm to about 165 nm; however, embodiments are not limited thereto.

[0057] The semiconductor light-emitting structure **110** may have a cylindrical shape having a center axis extending in the first direction. The semiconductor light-emitting structure **110** may have a width **110W** less than 100 μm in a second direction (e.g., X direction in FIG. 1) perpendicular to the first direction. In some embodiments, the width **110W** of the semiconductor light-emitting structure **110** may be about 100 nm to about 10 μm or about 500 nm to about 1,500 μm .

[0058] The transparent electrode layer **130** may have a width identical or similar to the width **110W** of the semiconductor light-emitting structure **110** in the second direction (e.g., X direction in FIG. 1). In the first direction, the transparent electrode layer **130** may have a variable thickness. In the transparent electrode layer **130**, a thickness of a portion in contact with the reflective electrode layer **170** in the first direction may be less than thicknesses of other portions of the transparent electrode layer **130**. In some embodiments, a maximum thickness of the transparent electrode layer **130** in the first direction may be about 50 nm to about 150 nm, and the thickness of the portion of the transparent electrode layer **130** in contact with the reflective electrode layer **170** may be about 30 nm to about 70 nm; however, embodiments are not limited thereto.

[0059] The transparent electrode layer **130** may include a transparent conductive material. In some embodiments, the transparent electrode layer **130** may include indium tin oxide (ITO), zinc-doped indium tin oxide (ZITO), zinc indium oxide (ZIO), gallium indium oxide (GIO), zinc tin oxide (ZTO), fluorine-doped tin oxide (FTO), aluminum-doped zinc oxide (AZO), gallium-doped zinc oxide (GZO), zinc magnesium oxide ($\text{In}_{0.4}\text{Sn}_{0.3}\text{O}_{1.2}$, $\text{Zn}_{1-x}\text{Mg}_x\text{O}$, $0 \leq x \leq 1$), or a combination thereof. In the second direction, the thickness of the transparent electrode layer **130** may be about 1 nm to about 100 nm, for example, about 7 nm to about 20 nm; however, embodiments are not limited thereto.

[0060] The reflective electrode layer **170** may include silver (Ag), nickel (Ni), aluminum (Al), chrome (Cr), rhodium (Rh), iridium (Ir), palladium (Pd), ruthenium (Ru), magnesium (Mg), zinc (Zn), platinum (Pt), gold (Au), titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), tungsten (W), or a combination thereof; however, embodiments are not limited thereto.

[0061] In a plan view (X-Y plane in FIG. 1), the semiconductor light-emitting structure **110** may have various plane shapes. For example, the plane shape of the semiconductor light-emitting structure **110** may be a circular shape, an elliptical shape, or a polygonal shape. The polygonal shape may be a tetragonal shape, a hexagonal shape, or an octagonal shape; however, embodiments are not limited thereto. In a plan view (X-Y plane in FIG. 1), a plane shape of the transparent electrode layer **130** and a plane shape of the semiconductor light-emitting structure **110** may correspond to each other. The semiconductor light-emitting structure **110** and the transparent electrode layer **130** may form a cylindrical shape.

[0062] A side wall of each of the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116** which are included in the semiconductor light-emitting structure **110** and a side wall of the transparent electrode layer **130** may be covered by a reflective structure **150**. In plan view (X-Y plane in FIG. 1), the semiconductor light-emitting structure **110** and the transparent electrode layer **130** may be surrounded by the reflective structure **150**. The reflective structure **150** may include a distributed Bragg reflector (DBR). As illustrated in FIG. 1, the reflective structure **150** may have a DBR structure in which a first insulating layer **150A**, a second insulating layer **150B**, a third insulating layer **150C**, and a fourth insulating layer **150D** are sequentially stacked. In this regard, the first insulating layer **150A** and the third insulating layer **150C** may include a first insulating material,

and the second insulating layer **150B** and the fourth insulating layer **150D** may include a second insulating material. The first insulating material and the second insulating material may have different refractive indexes from each other. In some embodiments, the first insulating material and the second insulating material may include different materials from each other, and the first insulating material and the second insulating material may each be selected from SiO₂, SiON, TiO₂, Si₃N₄, Al₂O₃, TiN, AlN, ZrO₂, TiAlN, TiSiN, HfO, NbO₂, TaO₂, and MgF₂.

[0063] The first insulating layer **150A** of the reflective structure **150** may be in contact with a side wall of each of the semiconductor light-emitting structure **110** and the transparent electrode layer **130**. The first insulating layer **150A** may include a material having strengthened total reflection characteristics. In some embodiments, the first insulating layer **150A** may include an insulating material having a refractive index less than or equal to a refractive index of the semiconductor light-emitting structure **110** and/or the transparent electrode layer **130**. For example, the first insulating layer **150A** may include SiO₂ or MgF₂; however, embodiments are not limited thereto. Each of the first insulating layer **150A**, the second insulating layer **150B**, the third insulating layer **150C**, and the fourth insulating layer **150D** of the reflective structure **150** may have a thickness of about 10 nm to about 200 nm.

[0064] The reflective structure **150** may reflect light traveling from the inside of the semiconductor light-emitting structure **110** to the side wall to adjust the light distribution. As the reflective structure **150** includes a DBR, the reflective structure **150** may act as a band pass filter (BPF) suppressing the transmission of light of a particular wavelength, and a transmittance difference according to an incident angle may effectively adjust the light distribution. In addition, the reflective structure **150** may cause a relatively significant increase in the intensity of light emitted from a particular region by using the transmittance difference according to an incident angle of light emitted from the semiconductor light-emitting structure **110**.

[0065] The light-emitting device **100** may include an insulating layer **160** covering the reflective structure **150**. In some embodiments, the insulating layer **160** may include a silicon oxide, a silicon nitride, or a combination thereof. For example, the insulating layer **160** may include tetraethyl ortho silicate (TEOS), undoped silicate glass (USG), phosphosilicate glass (PSG), borosilicate glass (BSG), borophosphosilicate glass (BPSG), fluoride silicate glass (FSG), spin on glass SOG), polysilazane, or a combination thereof.

[0066] The reflective electrode layer **170** may partially cover the insulating layer **160** and may be in contact with the transparent electrode layer **130** by penetrating the reflective structure **150** in the first direction. The reflective electrode layer **170** may include portions in contact with the insulating layer **160** and portions in contact with the reflective structure **150**.

[0067] According to the light-emitting device **100** described in relation to FIG. 1, by controlling the thickness **116T** of the second conductivity type semiconductor layer **116** according to a wavelength of light emitted from the semiconductor light-emitting structure **110**, the active layer **114** may be arranged at a position in the semiconductor light-emitting structure **110** to facilitate optimization of light extraction efficiency LEE of the light-emitting device **100**, and accordingly, the light extraction efficiency of the light-emitting device **100** may be maximized. In addition, as the semiconductor light-emitting structure **110** constitutes a micro LED having a width of about 100 μm or less, in the semiconductor light-emitting structure **110**, the active layer **114** may have a multi-quantum well structure for optimization of a micro-sized chip including the micro LED. Thus, according to some embodiments, the light-emitting device **100** having a structure better optimized to a micro-sized chip may be provided.

[0068] FIG. 2A is a schematic perspective view illustrating examples of shapes of the semiconductor light-emitting structure **110** and the transparent electrode layer **130**, and FIG. 2B is a schematic plan view illustrating the semiconductor light-emitting structure **110**.

[0069] Referring to FIGS. 2A and 2B, in a plan view (X-Y plane), each of the first conductivity

type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116**, which are included in the semiconductor light-emitting structure **110** and the transparent electrode layer **130** may have a tetragonal plane shape. A width **100W** of the semiconductor light-emitting structure **110** in a horizontal direction (e.g., X direction or Y direction) parallel with the main surface **102M** of the first conductivity type base semiconductor layer **102** (see FIG. 1) may be about 100 μm or less, about 50 μm or less, about 20 μm or less, about 10 μm or less, about 6 μm or less, about 5 μm or less, about 4 μm or less, or about 2 μm or less; however, embodiments are not limited thereto. In the horizontal direction (e.g., X direction or Y direction) parallel with the main surface **102M** of the first conductivity type base semiconductor layer **102**, the width of the transparent electrode layer **130** may be identical or similar to the width **100W** of the semiconductor light-emitting structure **110**.

[0070] In the transparent electrode layer **130**, a local recess **130R** may be formed in a surface opposite to a surface in contact with the semiconductor light-emitting structure **110**. A part of the reflective electrode layer **170** illustrated in FIG. 1 may be accommodated in the local recess **130R** of the transparent electrode layer **130**.

[0071] FIGS. 3A and 3B are diagrams illustrating a light-emitting device **100A** according to some embodiments. FIG. 3A is a schematic perspective view illustrating an example of a shape of the transparent electrode layer **130** and a semiconductor light-emitting structure **110A** included in the light-emitting device **100A**, and FIG. 3B is a schematic plan view illustrating an example of a plane shape of the semiconductor light-emitting structure **110A**. Like reference numerals in FIGS. 3A and 3B denote like members in FIG. 1, and thus any redundant description will be omitted.

[0072] Referring to FIGS. 3A and 3B, the light-emitting device **100A** may have substantially the same configuration as the light-emitting device **100** described in relation to FIG. 1, but may include the semiconductor light-emitting structure **110A** instead of the semiconductor light-emitting structure **110**. The semiconductor light-emitting structure **110A** may include a first conductivity type semiconductor layer **112A**, an active layer **114A**, and a second conductivity type semiconductor layer **116A**.

[0073] The first conductivity type semiconductor layer **112A**, the active layer **114A**, and the second conductivity type semiconductor layer **116A** may respectively have substantially the same configurations as the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116** described in relation to FIGS. 1, 2A, and 2B. In plan view (X-Y plane), the first conductivity type semiconductor layer **112A**, the active layer **114A**, and the second conductivity type semiconductor layer **116A** which are included in the semiconductor light-emitting structure **110A** may each have a circular plane shape.

[0074] FIGS. 4 and 5 are plan views illustrating a light-emitting device (**100B** and **100C**) according to some embodiments. FIG. 4 is a schematic plan view illustrating an example of a plane shape of a semiconductor light-emitting structure **110B** included in the light-emitting device **100B**, and FIG. 5 is a schematic plan view illustrating an example of a plane shape of a semiconductor light-emitting structure **110C** included in the light-emitting device **100C**.

[0075] Referring to FIG. 4, the light-emitting device **100B** may have substantially the same configuration as the light-emitting device **100** described in relation to FIG. 1, but may include the semiconductor light-emitting structure **110B** instead of the semiconductor light-emitting structure **110**. The semiconductor light-emitting structure **110B** may have substantially the same configuration as the semiconductor light-emitting structure **110** described in relation to FIG. 1. However, in plan view (X-Y plane), the semiconductor light-emitting structure **110B** may have a tetragonal plane shape with round edges.

[0076] Referring to FIG. 5, the light-emitting device **100C** may have substantially the same configuration as the light-emitting device **100** described in relation to FIG. 1, but may include the semiconductor light-emitting structure **110C** instead of the semiconductor light-emitting structure **110**. The semiconductor light-emitting structure **110C** may have substantially the same

configuration as the semiconductor light-emitting structure **110** described in relation to FIG. **1**. However, in plan view (X-Y plane), the semiconductor light-emitting structure **110C** may have a hexagonal plane shape.

[0077] FIGS. **6**, **7**, and **8** are cross-sectional views illustrating light-emitting devices (**100D**, **100E**, and **100F**) according to some embodiments. Like reference numerals in FIGS. **6**, **7**, and **8** denote like members in FIG. **1**, and thus any redundant description will be omitted.

[0078] Referring to FIG. **6**, the light-emitting device **100D** may have substantially the same configuration as the light-emitting device **100** described in relation to FIG. **1**, but may include a transparent electrode layer **130D** instead of the transparent electrode layer **130**.

[0079] The transparent electrode layer **130D** may have substantially the same configuration as the transparent electrode layer **130** described in relation to FIG. **1**. In the transparent electrode layer **130D**, a first surface **130D1** facing the second conductivity type semiconductor layer **116** and a second surface **130D2** facing the reflective electrode layer **170** may each extend flat in the horizontal direction (e.g., X direction and/or Y direction), and the entire transparent electrode layer **130D** may have a generally consistent thickness in the first direction (Z direction in FIG. **6**) (i.e., between the first surface **130D1** and the second surface **130D2**). In some embodiments, in the first direction (Z direction in FIG. **6**), the transparent electrode layer **130D** may have a thickness of about 50 nm to about 150 nm, for example, a thickness of about 100 nm; however, embodiments are not limited thereto.

[0080] Referring to FIG. **7**, the light-emitting device **100E** may have substantially the same configuration as the light-emitting device **100** described in relation to FIG. **1**, but may include an insulating spacer **120E** and a reflective structure **150E** instead of the reflective structure **150**. The insulating spacer **120E** and the reflective structure **150E** may each cover a side wall of each of the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116** which are included in the semiconductor light-emitting structure **110** in the horizontal direction parallel with the main surface **102M** of the first conductivity type base semiconductor layer **102**.

[0081] The insulating spacer **120E** may be in contact with the side wall of each of the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116** which are included in the semiconductor light-emitting structure **110**, as well as the side wall of the transparent electrode layer **130**. The reflective structure **150E** may be spaced apart from each of the semiconductor light-emitting structure **110** and the transparent electrode layer **130** with the insulating spacer **120E** arranged therebetween.

[0082] In some embodiments, the insulating spacer **120E** may include SiO.sub.2, SiON, TiO.sub.2, Si.sub.3N.sub.4, Al.sub.2O.sub.3, TiN, AlN, ZrO.sub.2, TiAlN, TiSiN, HfO, NbO.sub.2, TaO.sub.2, MgF.sub.2, or a combination thereof. The reflective structure **150E** may include a metal. For example, the reflective structure **150E** may include silver (Ag), nickel (Ni), aluminum (Al), chrome (Cr), rhodium (Rh), iridium (Ir), palladium (Pd), ruthenium (Ru), magnesium (Mg), zinc (Zn), platinum (Pt), gold (Au), titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), tungsten (W), or a combination thereof.

[0083] Referring to FIG. **8**, the light-emitting device **100F** may have substantially the same configuration as the light-emitting device **100** described in relation to FIG. **1**, but may further include a microlens **190**.

[0084] The microlens **190** may extract light emitted from the semiconductor light-emitting structure **110**. The microlens **190** may be spaced apart from the first conductivity type semiconductor layer **112** in the first direction (Z direction in FIG. **7**) with the first conductivity type base semiconductor layer **102** arranged therebetween. The main surface **102M** of the first conductivity type base semiconductor layer **102** may be in contact with the first conductivity type semiconductor layer **112**. The first conductivity type base semiconductor layer **102** may also include an emission surface **102E** including a part of a rear surface **102B** opposite to the main

surface **102M**. The microlens **190** may be arranged to overlap the semiconductor light-emitting structure **110** in the first direction. As the light-emitting device **100F** further includes the microlens **190**, the light extraction efficiency of the light-emitting device **100F** may be further improved. [0085] In some embodiments, the microlens **190** may include a spherical microlens or an aspherical microlens. In some embodiments, the microlens **190** may include a graded-refractive index layer formed in a multi-layer structure in which the refractive index gradually decreases in a light travel direction. The graded-refractive index layer may be formed by using an oblique deposition method, a sputtering method, an evaporation method, etc. The graded-refractive index layer may have a refractive index gradually decreasing in a direction towards a light output surface. In some embodiments, the microlens **190** may include TiO₂, SiC, GaN, GaP, SiN, SiON, ZrO₂, ITO, AlN, Al₂O₃, MgO, SiO₂, CaF₂, MgF₂, or a combination thereof; however, embodiments are not limited thereto.

[0086] FIG. **9A** is a cross-sectional view illustrating a light-emitting device **200** according to some embodiments. FIG. **9B** is an enlarged cross-sectional view illustrating a part denoted by “EX1.” Like reference numerals in FIGS. **9A** and **9B** denote like members in FIG. **1**, and thus any redundant description will be omitted.

[0087] Referring to FIGS. **9A** and **9B**, the light-emitting device **200** may have substantially the same configuration as the light-emitting device **100** described in relation to FIG. **1**, but may include a semiconductor light-emitting structure **210**.

[0088] The semiconductor light-emitting structure **210** may have substantially the same configuration as the semiconductor light-emitting structure **110** described in relation to FIG. **1**, but may include an active layer **214**. The active layer **214** may include a first multi-quantum well layer **214A**, a spacer barrier layer **214B**, and a second multi-quantum well layer **214C**, which are sequentially stacked on the first conductivity type semiconductor layer **112**.

[0089] In the active layer **214**, the first multi-quantum well layer **214A** may include a surface in contact with the first conductivity type semiconductor layer **112**, and the second multi-quantum well layer **214C** may include a surface in contact with the second conductivity type semiconductor layer **116**. The spacer barrier layer **214B** may be arranged between the first multi-quantum well layer **214A** and the second multi-quantum well layer **214C**. The first multi-quantum well layer **214A** and the second multi-quantum well layer **214C** may be spaced apart from each other in the first direction (Z direction in FIGS. **9A** and **9B**) with the spacer barrier layer **214B** arranged therebetween.

[0090] In the active layer **214**, the first multi-quantum well layer **214A** and the second multi-quantum well layer **214C** may each have substantially the same configuration as the active layer **114** described in relation to FIG. **1**. The first multi-quantum well layer **214A** and the second multi-quantum well layer **214C** may each have a thickness less than about 300 nm in the first direction. In the first direction, a thickness BT of the spacer barrier layer **214B** may be greater than a thickness AT of the first multi-quantum well layer **214A** and a thickness CT of the second multi-quantum well layer **214C**.

[0091] In the active layer **214**, at least one constituent material of the spacer barrier layer **214B** may be identical to at least one constituent material selected in the first multi-quantum well layer **214A** and the second multi-quantum well layer **214C**. In some embodiments, the first multi-quantum well layer **214A** and the second multi-quantum well layer **214C** may each have a multi-quantum well structure in which a plurality of quantum barrier layers and a plurality of quantum well layers are alternately arranged. For example, in each of the first multi-quantum well layer **214A** and the second multi-quantum well layer **214C**, the quantum well layer may include an undoped In_xGa_{1-x}N layer (0<x<1), and the quantum barrier layer may include an undoped GaN layer or a GaN layer doped with silicon (Si). In some embodiments, the spacer barrier layer **214B** may include an In_xGa_{1-x}N layer (0<x<1), an undoped GaN layer, or a GaN layer doped with silicon (Si).

[0092] FIG. 10 is a diagram illustrating a relation between a change in the light extraction efficiency LEE according to a position of the active layer **214** of the semiconductor light-emitting structure **210** included in the light-emitting device **200** illustrated in FIGS. 9A and 9B and a position of each of the first multi-quantum well layer **214A** and the second multi-quantum well layer **214C** included in the active layer **214** of the semiconductor light-emitting structure **210**. [0093] Referring to FIG. 10, in the active layer **214**, each of the second multi-quantum well layer **214C** and the first multi-quantum well layer **214A** may be arranged at a position corresponding to a peak (PK1 and PK2) of a value of the light extraction efficiency LEE of light passing through the active layer **214** of the semiconductor light-emitting structure **210**. The second multi-quantum well layer **214C** may extend from an interface between the second conductivity type semiconductor layer **116** and the second multi-quantum well layer **214C** to a position passing the first peak PK1 of the value of the light extraction efficiency LEE in the first direction (Z direction). The first multi-quantum well layer **214A** may extend from an interface between the spacer barrier layer **214B** and the first multi-quantum well layer **214A** to a position passing the second peak PK2 of the value of the light extraction efficiency LEE in the first direction (Z direction).

[0094] Each of the thickness AT of the first multi-quantum well layer **214A** and the thickness CT of the second multi-quantum well layer **214C** may be less than the thickness BT of the spacer barrier layer **214B**, and the thickness BT of the spacer barrier layer **214B** may be less than a distance between the two adjacent peaks (PK1 and PK2) of the value of the light extraction efficiency LEE of light passing through the active layer **214**. Each of the thickness AT of the first multi-quantum well layer **214A** and the thickness CT of the second multi-quantum well layer **214C** may be less than 300 nm, less than 200 nm, less than 100 nm, less than 50 nm, less than 40 nm, less than 20 nm, less than 100, less than 5 nm, or less than 3 nm. For example, the thickness **114T** of the active layer **114** may be within a range from about 2 nm to about 10 nm; however, embodiments are not limited thereto. In some embodiments, each of the thickness AT of the first multi-quantum well layer **214A** and the thickness CT of the second multi-quantum well layer **214C** may be less than the thickness of a single multi-quantum well layer when the active layer **214** includes a single multi-quantum well layer.

[0095] FIGS. 11 to 13 are diagrams illustrating a display device **400** according to some embodiments. FIG. 11 is a schematic perspective view of the display device **400**, FIG. 12 is an enlarged plan view of a part denoted by "EX2" in FIG. 11, and FIG. 13 is a cross-sectional view schematically illustrating components of a portion along a cross-section cut by line I-I' in FIG. 11 and a portion along a cross-section cut by line II-II' in FIG. 12. Like reference numerals in FIGS. 11 to 13 denote like members in FIG. 1, and thus any redundant description will be omitted.

[0096] Referring to FIGS. 11 to 13, the display device **400** may include a pixel array **410** and a circuit board **420** which are arranged to overlap each other in the first direction (Z direction in FIG. 11). The circuit board **420** may include driver circuits. The pixel array **410** may include a plurality of pixels PX arranged in a pixel region PXR on the circuit board **420**. The display device **400** may further include a frame **402** surrounding the pixel array **410** and the circuit board **420**.

[0097] The circuit board **420** may be a driver circuit board including a plurality of transistors. In some embodiments, the circuit board **420** may include an application-specific integrated circuit (ASIC) including a plurality of driver circuits. In some embodiments, the circuit board **420** may include a flexible substrate. In this case, the display device **400** may be implemented as a variable or curved display device.

[0098] The pixel array **410** may include the pixel region PXR in which the plurality of pixels PX, a plurality of contact pad regions PAD in which a contact pad electrode **494** is arranged, a connection region CR for interconnecting the pixels PX to the contact pad electrode **494**, and an edge region ISO.

[0099] The plurality of pixels PX may include a plurality of first subpixels SP1, a plurality of second subpixels SP2, and a plurality of third subpixels SP3, which are configured to emit light of

a particular wavelength, for example, light of a particular color. The plurality of first subpixels SP1, the plurality of second subpixels SP2, and the plurality of third subpixels SP3 may each include a light-emitting device having the configuration described in relation to the semiconductor light-emitting structure **110** with reference to FIG. **1**.

[0100] In some embodiments, the first to third subpixels SP1, SP2, and SP3 may respectively be configured to emit red (R) light, green (G) light, and blue (B) light. In some embodiments, each of the plurality of pixels PX may include the first to third subpixels SP1, SP2, and SP3 arranged in a Bayer pattern. That is, each of the plurality of pixels PX may include the first and third subpixels SP1 and SP3 arranged in a first diagonal direction and two second subpixels SP2 arranged in a second diagonal direction intersecting with the first diagonal direction. FIG. **12** illustrates that the first to third subpixels SP1, SP2, and SP3 are arranged in a 2×2 Bayer pattern in each of the plurality of pixels PX; however, embodiments are not limited thereto. For example, each of the plurality of pixels PX may have other arrangements, such as 3×3, 4×4, etc. In some embodiments, some of the plurality of pixels PX may be configured to emit light having a color other than red (R), green (G), and blue (B), for example, yellow light. FIG. **11** illustrates that in the pixel array **410**, the plurality of pixels PX are in a 15×15 arrangement along a column direction and a row direction; however, embodiments are not limited thereto. The pixel array **410** may include a different number of pixels PX along a column direction and a row direction, for example, a plurality of pixels PX having a 1,024×768 arrangement.

[0101] The plurality of contact pad regions PAD may be arranged along an edge of the display device **400** on at least one side of the pixel region PXR. The plurality of contact pad regions PAD may be electrically connected to the plurality of pixels PX and the driver circuits of the circuit board **420**. The display device **400** may be electrically connected to an external device through the plurality of contact pad regions PAD. The number of the contact pad regions PAD included in the display device **400** may vary. In some embodiments, the number of the plurality of contact pad regions PAD included in the display device **400** may be determined according to the number of the pixels PX included in the pixel array **410**, a driving method of the driver circuit included in the circuit board **420**, etc.

[0102] The connection region CR may be arranged between the pixel region PXR and the plurality of contact pad regions PAD. In the connection region CR, a wiring structure electrically connected to the plurality of pixels PX, for example, a part of a grid electrode **492** and a common electrode **445** as illustrated in FIG. **13**.

[0103] The edge region ISO of the display device **400** may be a region along edges of the pixel array **410**. The semiconductor light-emitting structure **110** may not be arranged in the edge region ISO.

[0104] The frame **402** of the display device **400** may be arranged around the pixel array **410** and may serve as a guide defining a space in which the pixel array **410** is arranged. The frame **402** may include polymer, ceramic, semiconductor, metal, or a combination thereof.

[0105] As illustrated in FIG. **13**, the circuit board **420** may include a semiconductor substrate **422**, a driver circuit formed on the semiconductor substrate **422** and including a plurality of driver devices **424** including transistors, a plurality of interconnections **426** electrically connected to the plurality of driver devices **424**, and a plurality of wiring lines **430** connected to the plurality of interconnections **426**. The plurality of driver devices **424** constituting the driver circuit, the plurality of interconnections **426**, and the plurality of wiring lines **430** may be covered by an insulating layer **428**. The circuit board **420** may further include a first bonding insulating layer **440** on the insulating layer **428** and a plurality of first bonding electrodes **442** penetrating the first bonding insulating layer **440** and connected to the plurality of wiring lines **430**.

[0106] The semiconductor substrate **422** may include a plurality of impurity regions **432** constituting source region/drain regions of the plurality of transistors constituting the plurality of driver devices **424**. The semiconductor substrate **422** may include a semiconductor such as silicon

(Si) or germanium (Ge) or a compound semiconductor such as SiGe, SiC, GaAs, InAs, or InP. The semiconductor substrate **422** may further include a plurality of through electrodes **450** connected to the driver circuit, such as a through silicon via (TSV) and a plurality of substrate wiring lines **452** connected to the plurality of through electrodes **450**.

[0107] The driver circuit may be a circuit for controlling the drive of the pixels PX or the first to third subpixels SP1, SP2, and SP3. Some of the plurality of impurity regions **432** may be electrically connected to at least one of the first to third subpixels SP1, SP2, and SP3 through the interconnections **426**, the wiring lines **430**, and the first bonding electrodes **442**. In some embodiments, some of the plurality of impurity regions **432** may be connected to one substrate wiring line **452** from among the plurality of substrate wiring lines **452** through the through electrode **450**.

[0108] Upper surfaces of the plurality of first bonding electrodes **442** and an upper surface of the first bonding insulating layer **440** may constitute an upper surface of the circuit board **420**. The plurality of first bonding electrodes **442** included in the circuit board **420** may be bonded to a plurality of second bonding electrodes **176** included in the pixel array **410** and provide an electrical connection path. In some embodiments, the plurality of first bonding electrodes **442** and the plurality of second bonding electrodes **176** may each include a copper (Cu) film. Each of the plurality of first bonding electrodes **442** and the plurality of second bonding electrodes **176** may further include a barrier metal layer surrounding the copper (Cu) film. The barrier metal layer may include Ta, TaN, or a combination thereof.

[0109] The first bonding insulating layer **440** included in the circuit board **420** may be bonded to a second bonding insulating layer **162** included in the pixel array **410**. Each of the first bonding insulating layer **440** and the second bonding insulating layer **162** may include SiO, SiN, SiCN, SiOC, SiON, SiOCN, or a combination thereof.

[0110] In the pixel array **410**, each of the first to third subpixels SP1, SP2, and SP3 may include the semiconductor light-emitting structure **110** described in relation to FIG. 1. More specifically, the pixel array **410** may include the first conductivity type base semiconductor layer **102** including a main surface **110M** and the rear surface **102B** which face each other and the plurality of semiconductor light-emitting structures **110** arranged on the main surface **102M** of the first conductivity type base semiconductor layer **102**. The plurality of semiconductor light-emitting structures **110** may be spaced apart from each other in the horizontal direction parallel with the main surface **110M** of the first conductivity type base semiconductor layer **102**. Each of the plurality of semiconductor light-emitting structures **110** may include the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116** which are sequentially stacked in the first direction perpendicular to the main surface **110M** of the first conductivity type base semiconductor layer **102**. Detailed configurations of the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116** are each the same as described above in relation to FIG. 1.

[0111] The pixel array **410** may further include the grid electrode **492** penetrating the first conductivity type base semiconductor layer **102** in the first direction. The grid electrode **492** may include a metal layer having a grid shape. The grid electrode **492** may be referred to as a first electrode. The grid electrode **492** may include local regions extending along regions between the plurality of semiconductor light-emitting structures **110** in a plan view parallel with the main surface **102M** of the first conductivity type base semiconductor layer **102**, and the local regions of the grid electrode **492** may be connected to each other to form a single layer and may be arranged to surround each of the plurality of semiconductor light-emitting structures **110**. The local regions of the grid electrode **492** may be connected to each other to form a grid shape or a mesh shape. As such, as the grid electrode **492** is arranged to fill the spaces between the plurality of semiconductor light-emitting structures **110**, the spreading of current may be improved, and the light emission efficiency at the pixel array **410** of the display device **400** may increase. In some embodiments, by

forming the grid electrode **492** through a plating process, the relatively narrow spaces between the plurality of semiconductor light-emitting structures **110** may be stably filled.

[0112] A part of the grid electrode **492** may be in contact with a side wall of the first conductivity type base semiconductor layer **102**. Another part of the grid electrode **492** may be in contact with the rear surface **102B** of the first conductivity type base semiconductor layer **102** to define the emission surface **102E** including a part of the rear surface **102B** of the first conductivity type base semiconductor layer **102**. In some embodiments, the grid electrode **492** may include silver (Ag), nickel (Ni), aluminum (Al), chrome (Cr), rhodium (Rh), iridium (Ir), palladium (Pd), ruthenium (Ru), magnesium (Mg), zinc (Zn), platinum (Pt), gold (Au), or a combination thereof.

[0113] A plurality of microlenses **496** may be arranged on the emission surface **102E** defined by the grid electrode **492**, of the rear surface **102B** of the first conductivity type base semiconductor layer **102**. Each of the plurality of microlenses **496** may be in contact with a portion covering the rear surface **102B**, of the grid electrode **492**. The plurality of microlenses **496** may be arranged to overlap the plurality of semiconductor light-emitting structures **110** in the direction perpendicular to the main surface **102M** of the first conductivity type base semiconductor layer **102**. Detailed configuration of the plurality of microlenses **496** is substantially the same as the configuration of the microlens **190** described in relation to FIG. 8.

[0114] The pixel array **410** of the display device **400** may further include the transparent electrode layer **130**, the reflective structure **150**, and the reflective electrode layer **170** penetrating the reflective structure **150** and in contact with the transparent electrode layer **130**. Detailed configurations of the transparent electrode layer **130**, the reflective structure **150**, and the reflective electrode layer **170** are the same as described in relation to FIG. 1.

[0115] The plurality of semiconductor light-emitting structures **110** included in the pixel array **410** of the display device **400** may be configured to emit light having a wavelength (λ) within a range from about 400 nm to about 700 nm, for example, a wavelength (λ) within a range from about 490 nm to about 700 nm. A minimum distance from the transparent electrode layer **130** to the active layer **114** in the direction perpendicular to the main surface **102M** of the first conductivity type base semiconductor layer **102** may be determined according to wavelength (λ) of light emitted from the semiconductor light-emitting structure **110**. A minimum distance from the reflective electrode layer **170** to the active layer **114** in the direction perpendicular to the main surface **102M** of the first conductivity type base semiconductor layer **102** may be greater than 0.05λ and less than 0.4λ . In some embodiments, the minimum distance from the reflective electrode layer **170** to the active layer **114** in the direction perpendicular to the main surface **102M** of the first conductivity type base semiconductor layer **102** may be greater than 0.05λ and less than 0.24λ .

[0116] In the pixel array **410** of the display device **400**, the plurality of second bonding electrodes **176** may be connected to the plurality of reflective electrode layers **170**, the second bonding insulating layer **162** surrounding the plurality of second bonding electrodes **176** may be in contact with a part of the reflective electrode layer **170** and may be in contact with the insulating layer **160** covering the reflective structure **150**.

[0117] The display device **400** may further include a common electrode **145** and an inner pad electrode **447**. In the pixel array **410**, the insulating layer **160** covering the reflective electrode layer **170** may extend to the connection region CR and the contact pad regions PAD to cover the common electrode **145** and the inner pad electrode **447**.

[0118] The grid electrode **492** may extend from the pixel region PXR to the connection region CR and may be physically in contact with the first conductivity type base semiconductor layer **102** and the common electrode **445** in the connection region CR. The grid electrode **492** may be electrically connected to the first conductivity type base semiconductor layer **102** and the common electrode **445**. In the contact pad regions PAD, the contact pad electrode **494** may be arranged on the inner pad electrode **447**.

[0119] The plurality of second bonding electrodes **176** may be connected to the common electrode

445. The common electrode **445** may have a ring shape or a tetragonal ring shape surrounding a pixel region PXA in a plan view parallel with the main surface **102M** of the first conductivity type base semiconductor layer **102**. However, the arrangement of the common electrode **445** may be variously modified or changed according to a need.

[0120] In the contact pad regions PAD, the contact pad electrode **494** may be arranged on the inner pad electrode **447**. The inner pad electrode **447** may be in contact with the contact pad electrode **494**. The inner pad electrode **447** may be arranged between the contact pad electrode **494** and the second bonding electrodes **176** and interconnect the contact pad electrode **494** and the second bonding electrodes **176** to each other. The common electrode **445** and the inner pad electrode **447** may include a conductive material, for example, silver (Ag), nickel (Ni), aluminum (Al), chrome (Cr), rhodium (Rh), iridium (Ir), palladium (Pd), ruthenium (Ru), magnesium (Mg), zinc (Zn), platinum (Pt), gold (Au), or a combination thereof.

[0121] The contact pad electrode **494** may be connected to an external device or an external integrated circuit (IC) which may apply an electric signal to the circuit board **420** through wire bonding or anisotropic conductive film (AFC) bonding. The contact pad electrode **494** may electrically connect the external device to the driver circuits of the circuit board **420**. The contact pad electrode **494** may include a metal, for example, gold (Au), silver (Ag), nickel (Ni), etc.

[0122] From among the plurality of second bonding electrodes **176**, the second bonding electrode **176** arranged in the pixel region PXR may be connected to the reflective electrode layer **170**, from among the plurality of second bonding electrodes **176**, the second bonding electrode **176** arranged in the connection region CR may be connected to the common electrode **445**, and from among the second bonding electrodes **176**, the second bonding electrode **176** arranged in the contact pad region PAD may be connected to the inner pad electrode **447**. The grid electrode **492** may be connected to the plurality of second bonding electrodes **176** through the common electrode **445**.

[0123] A surface of the second bonding insulating layer **162** which faces the circuit board **420** and surfaces of the plurality of second bonding electrodes **176** which face the circuit board **420** may extend on one plane. The second bonding insulating layer **162** may form a dielectric-dielectric bonding along with the first bonding insulating layer **440**. The circuit board **420** and the pixel array **410** may be bonded to each other by the bonding between the plurality of first bonding electrodes **442** and the plurality of second bonding electrodes **176** and the bonding between the first bonding insulating layer **440** and the second bonding insulating layer **162**.

[0124] In some embodiments, the bonding between the plurality of first bonding electrodes **442** and the plurality of second bonding electrodes **176** may be, for example, copper (Cu)-copper (Cu) bonding, and the bonding between the first bonding insulating layer **440** and the second bonding insulating layer **162** may be dielectric-dielectric bonding such as SiCN—SiCN bonding. The circuit board **420** and the pixel array **410** may be bonded by hybrid bonding including copper (Cu)-copper (Cu) bonding and dielectric-dielectric bonding or may be bonded without a separate bonding layer.

[0125] The display device **400** may include the plurality of semiconductor light-emitting structures **110**, and by controlling the thickness **116T** of the second conductivity type semiconductor layer **116** according to a wavelength of light emitted from the plurality of semiconductor light-emitting structures **110**, the active layer **114** may be arranged in a position facilitating the optimization of light extraction efficiency LEE of the light-emitting device **100** in each of the plurality of semiconductor light-emitting structures **110**. Accordingly, the light extraction efficiency may be maximized in the pixel region PXA of the display device **400**. In addition, in the pixel region PXA of the display device **400**, the semiconductor light-emitting structure **110** may constitute a micro LED having a width of about 100 μm or less, and in the semiconductor light-emitting structure **110**, the active layer **114** may have a multi-quantum well structure optimized to a micro-sized chip including the micro LED. Accordingly, the display device **400** having a structure optimized to a micro-sized chip may be provided.

[0126] FIGS. **14** and **15** are graphs showing simulation results for evaluation of the light extraction

efficiency LEE of a light-emitting device according to some embodiments.

[0127] For evaluation of FIG. 14, the light extraction efficiency LEE according to the thickness **116T** of the second conductivity type semiconductor layer **116** was evaluated for each of red light having a wavelength (λ) of 630 nm, green light having a wavelength (λ) of 530 nm, and blue light having a wavelength (λ) of 460 nm in the light-emitting device including the semiconductor light-emitting structure **110** having the configuration described in relation to FIG. 1. In FIG. 14, the vertical axis represents the light extraction efficiency LEE within an emission angle range of ± 10 degrees.

[0128] From the evaluation result of FIG. 14, a range of the thickness **116T** of the second conductivity type semiconductor layer **116** may be determined based on a position where a peak value from among the values of light extraction efficiency LEE is present in each of red light, green light, and blue light.

[0129] In some embodiments, based on the result of FIG. 14, for the red light having a wavelength (λ) of 630 nm, the thickness **116T** of the second conductivity type semiconductor layer **116** may be set to be greater than 0.05λ and less than 0.3λ . For example, when the thickness **116T** of the second conductivity type semiconductor layer **116** is set to be greater than 0.15λ and less than 0.21λ , the thickness **116T** of the second conductivity type semiconductor layer **116** may be within a range from about 95 nm to about 135 nm.

[0130] In some embodiments, based on the result of FIG. 14, for the green light having a wavelength (λ) of 530 nm, the thickness **116T** of the second conductivity type semiconductor layer **116** may be set to be greater than 0.05λ and less than 0.3λ . For example, when the thickness **116T** of the second conductivity type semiconductor layer **116** is set to be greater than 0.11λ and less than 0.19λ , the thickness **116T** of the second conductivity type semiconductor layer **116** may be within a range from about 60 nm to about 100 nm.

[0131] In some embodiments, based on the result of FIG. 14, for the blue light having a wavelength (λ) of 460 nm, the thickness **116T** of the second conductivity type semiconductor layer **116** may be set to be greater than 0.05λ and less than 0.4λ . For example, when the thickness **116T** of the second conductivity type semiconductor layer **116** is set to be greater than 0.27λ and less than 0.36λ , the thickness **116T** of the second conductivity type semiconductor layer **116** may be within a range from about 125 nm to about 165 nm.

[0132] For the evaluation of FIG. 15, a method substantially the same as the evaluation method of FIG. 14 was used, but the vertical axis in FIG. 15 represents the light extraction efficiency LEE without limiting an emission angle to a particular range.

[0133] From the evaluation result of FIG. 15, a range of the thickness **116T** of the second conductivity type semiconductor layer **116** may be determined based on a position where a peak value from among the values of light extraction efficiency LEE is present in each of red light, green light, and blue light.

[0134] In some embodiments, based on the result of FIG. 15, for the red light having a wavelength (λ) of 630 nm, the thickness **116T** of the second conductivity type semiconductor layer **116** may be set to be greater than 0.05λ and less than 0.3λ . For example, when the thickness **116T** of the second conductivity type semiconductor layer **116** is set to be greater than 0.18λ and less than 0.25λ , the thickness **116T** of the second conductivity type semiconductor layer **116** may be within a range from about 115 nm to about 155 nm.

[0135] In some embodiments, based on the result of FIG. 15, for the green light having a wavelength (λ) of 530 nm, the thickness **116T** of the second conductivity type semiconductor layer **116** may be set to be greater than 0.05λ and less than 0.3λ . For example, when the thickness **116T** of the second conductivity type semiconductor layer **116** is set to be greater than 0.12λ and less than 0.2λ , the thickness **116T** of the second conductivity type semiconductor layer **116** may be within a range from about 65 nm to about 105 nm.

[0136] In some embodiments, based on the result of FIG. 15, for the blue light having a wavelength

(λ) of 460 nm, the thickness **116T** of the second conductivity type semiconductor layer **116** may be set to be greater than 0.05λ and less than 0.4λ . For example, when the thickness **116T** of the second conductivity type semiconductor layer **116** is set to be greater than 0.30λ and less than 0.39λ , the thickness **116T** of the second conductivity type semiconductor layer **116** may be within a range from about 140 nm to about 180 nm.

[0137] FIGS. **16**, **17**, and **18** illustrate graphs showing simulation results for evaluation of light extraction efficiency LEE of a light-emitting device according to some embodiments.

[0138] For evaluation of FIGS. **16**, **17**, and **18**, the light extraction efficiency LEE according to the thickness **116T** of the second conductivity type semiconductor layer **116** was evaluated for each of red light having a wavelength (λ) of 630 nm, green light having a wavelength (λ) of 530 nm, and blue light having a wavelength (λ) of 460 nm in the light-emitting device including the semiconductor light-emitting structure **210** having the configuration described in relation to FIGS. **9A** and **9B**. In FIGS. **16**, **17**, and **18**, the vertical axis represents the light extraction efficiency LEE within an emission angle range of ± 10 degrees.

[0139] From the evaluation results of FIGS. **16**, **17**, and **18**, a range of the thickness **116T** of the second conductivity type semiconductor layer **116** may be determined based on a position where a peak value from among the values of light extraction efficiency LEE is present in each of red light, green light, and blue light, according to a method similar to the method described in relation to FIG. **14**. In addition, based on the evaluation results of FIGS. **16**, **17**, and **18**, the second multi-quantum well layer **214C** and the first multi-quantum well layer **214A** in the active layer **214** may respectively be designed to be located at positions corresponding to the first peak and the second peak of the light extraction efficiency LEE value of light passing through the active layer **214** of the semiconductor light-emitting structure **210**.

[0140] For example, as for red light, the size of the active layer **214** may be designed based on the evaluation result of FIG. **16**. That is, a thickness **CTR1** of the second multi-quantum well layer **214C** may be designed to extend from the interface between the second conductivity type semiconductor layer **116** and the second multi-quantum well layer **214C** to a position passing the first peak of the value of the light extraction efficiency LEE towards the first conductivity type semiconductor layer **112**. A thickness **CTR2** of the first multi-quantum well layer **214A** may be designed to extend from the interface between the spacer barrier layer **214B** and the first multi-quantum well layer **214A** to a position passing the second peak of the value of the light extraction efficiency LEE towards the first conductivity type semiconductor layer **112**. A thickness **BT1** of the spacer barrier layer **214B** may be designed to be greater than the thickness **CTR1** of the first multi-quantum well layer **214A**, greater than the thickness **CTR2** of the second multi-quantum well layer **214C**, and less than a distance **TRI** between a position where the first peak of the light extraction efficiency LEE value is present and a position where the second peak of the light extraction efficiency LEE value is present.

[0141] As for green light, the size of the active layer **214** may be designed based on the evaluation result of FIG. **17**. That is, a thickness **CTG1** of the second multi-quantum well layer **214C** may be designed to extend from the interface between the second conductivity type semiconductor layer **116** and the second multi-quantum well layer **214C** to a position passing the first peak of the value of the light extraction efficiency LEE towards the first conductivity type semiconductor layer **112**. A thickness **CTG2** of the first multi-quantum well layer **214A** may be designed to extend from the interface between the spacer barrier layer **214B** and the first multi-quantum well layer **214A** to a position passing the second peak of the value of the light extraction efficiency LEE towards the first conductivity type semiconductor layer **112**. A thickness **BT2** of the spacer barrier layer **214B** may be designed to be greater than the thickness **CTG1** of the first multi-quantum well layer **214A**, greater than the thickness **CTG2** of the second multi-quantum well layer **214C**, and less than a distance **TG1** between a position where the first peak of the light extraction efficiency LEE value is present and a position where the second peak of the light extraction efficiency LEE value is

present.

[0142] As for blue light, the size of the active layer **214** may be designed based on the evaluation result of FIG. **18**. That is, a thickness CTB1 of the second multi-quantum well layer **214C** may be designed to extend from the interface between the second conductivity type semiconductor layer **116** and the second multi-quantum well layer **214C** to a position passing the first peak of the value of the light extraction efficiency LEE towards the first conductivity type semiconductor layer **112**. A thickness CTB2 of the first multi-quantum well layer **214A** may be designed to extend from the interface between the spacer barrier layer **214B** and the first multi-quantum well layer **214A** to a position passing the second peak of the value of the light extraction efficiency LEE towards the first conductivity type semiconductor layer **112**. A thickness BT3 of the spacer barrier layer **214B** may be designed to be greater than the thickness CTB1 of the first multi-quantum well layer **214A**, greater than the thickness CTB2 of the second multi-quantum well layer **214C**, and less than a distance TB1 between a position where the first peak of the light extraction efficiency LEE value is present and a position where the second peak of the light extraction efficiency LEE value is present.

[0143] FIGS. **19A** to **19G** each illustrate a cross-sectional view illustrating a method of manufacturing an display device including a light-emitting device according to a process order, according some embodiments. Referring to FIGS. **19A** to **19G**, an example of a method of manufacturing the display device **400** described in relation to FIGS. **11**, **12**, and **13** is described. Like reference numerals in FIGS. **19A** to **19G** denote like members in FIG. **1** to **13**, and thus any redundant description will be omitted.

[0144] Referring to FIG. **19A**, by using a semiconductor single crystal growth process, a deposition process, an etching process, etc. using a substrate for growth **401**, a structure in which the first conductivity type base semiconductor layer **102**, the plurality of semiconductor light-emitting structures **110** arranged in the pixel region PXR and each including the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116**, and the plurality of transparent electrode layers **130** covering the plurality of semiconductor light-emitting structures **110** are arranged may be formed on the substrate for growth **401**.

[0145] The substrate for growth **401** may be a substrate for semiconductor single crystal growth and may include AlN, AlGa_N, ZnO, GaAs, MgAl_{sub.2}O_{sub.4}, MgO, LiAlO_{sub.2}, LiGaO_{sub.2}, GaN, or a combination thereof. In some embodiments, for improvement in crystallinity of and semiconductor layers and light extraction efficiency, at least a part of an upper surface of the substrate for growth **401** may have an uneven structure. In this case, layers grown on the substrate for growth **401** may also have an uneven structure.

[0146] In some embodiments, to form the structure illustrated in FIG. **19A**, the first conductivity type base semiconductor layer **102**, the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116** may be sequentially formed on the substrate for growth **401**, and the transparent electrode layer **130** may be formed on the second conductivity type semiconductor layer **116**. By an etching process using a hard mask pattern as an etching mask, each of the transparent electrode layer **130**, the second conductivity type semiconductor layer **116**, the active layer **114**, and the first conductivity type semiconductor layer **112** may be partially etched such that the plurality of semiconductor light-emitting structures **110** and the plurality of transparent electrode layers **130** covering the plurality of semiconductor light-emitting structures **110** which are spaced apart from each other are left on the first conductivity type base semiconductor layer **102**. The plurality of semiconductor light-emitting structures **110** may form a plurality of cylindrical shapes having a circular, elliptical, or polygonal plane shape, along with the plurality of transparent electrode layers **130**.

[0147] The first conductivity type base semiconductor layer **102**, the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor

layer **116** may be formed by using a metal organic chemical vapor deposition (MOCVD) process, a hydride vapor phase epitaxy (HVPE) process, or a molecular beam epitaxy (MBE) process. The constituent materials and thicknesses of the first conductivity type base semiconductor layer **102**, the first conductivity type semiconductor layer **112**, the active layer **114**, and the second conductivity type semiconductor layer **116** may be the same as described in relation to FIG. **1**. For example, the second conductivity type semiconductor layer **116** in different semiconductor light-emitting structures **110** may have different thicknesses.

[0148] In some embodiments, in each of the plurality of semiconductor light-emitting structures **110**, a wet etching process may be further performed to remove damaged regions due to the etching. In the wet etching process, by controlling process conditions to differentiate the selectivity of crystal surfaces, the damaged portions may be selectively removed from each of the plurality of semiconductor light-emitting structure **110**, and the side wall of each of the plurality of semiconductor light-emitting structures **110** may have a profile extending in a direction perpendicular to the main surface **102M** of the first conductivity type base semiconductor layer **102**. In addition, the non-radiative recombination due to the damaged regions may decrease in the side wall of each of the plurality of semiconductor light-emitting structures **110**, which may lead to improved luminance in a light-emitting device.

[0149] Referring to FIG. **19B**, from the resulting product of FIG. **19A**, the reflective structure **150** covering surfaces of the first conductivity type base semiconductor layer **102**, the plurality of semiconductor light-emitting structures **110**, and the plurality of transparent electrode layers **130** may be formed, and the reflective structure **150** may be partially removed in the connection region CR and the contact pad region PAD to expose the first conductivity type base semiconductor layer **102**. By partially removing the exposed first conductivity type base semiconductor layer **102** by a certain thickness, the thickness of the first conductivity type base semiconductor layer **102** may be reduced in the connection region CR and the contact pad region PAD.

[0150] Referring to FIG. **19C**, the common electrode **445** arranged on the first conductivity type base semiconductor layer **102** may be formed in the connection region CR, and the inner pad electrode **447** arranged on the first conductivity type base semiconductor layer **102** may be formed in the contact pad region PAD. The insulating layer **160** covering a resulting product may be formed. The insulating layer **160** may be formed to cover the reflective structure **150** in the pixel region PXR and to cover the common electrode **445** and the inner pad electrode **447** in the connection region CR and the contact pad region PAD. The insulating layer **160** may be formed to have a flat upper surface.

[0151] Referring to FIG. **19D**, from the resulting product of FIG. **19C**, parts of the insulating layer **160** and the reflective structure **150** may be etched to expose the plurality of transparent electrode layers **130** in the pixel region PXR. In this regard, by overetching, the exposed parts of the plurality of transparent electrode layers **130** may be partially etched to form the local recess **130R** (see FIG. **2A**) in the exposed surface of each of the plurality of transparent electrode layers **130**.

[0152] The plurality of reflective electrode layers **170** covering the local recess **130R** of each of the plurality of transparent electrode layers **130** and the peripheral reflective structure **150** and insulating layer **160** may be formed. Each of the plurality of reflective electrode layers **170** may be formed to conformally cover surfaces defining the local recess **130R** of the transparent electrode layer **130** and surfaces of the reflective structure **150** and the insulating layer **160** which are adjacent to the local recess **130R**.

[0153] The second bonding insulating layer **162** covering the plurality of reflective electrode layers **170** and insulating layer **160** in the pixel region PXR and covering the insulating layer **160** in the connection region CR and the contact pad region PAD may be formed. The second bonding insulating layer **162** may be formed to have a flat upper surface.

[0154] Referring to FIG. **19E**, by partially etching the second bonding insulating layer **162** in the pixel region PXR, the connection region CR, and the contact pad region PAD, a plurality of via

holes exposing the plurality of reflective electrode layers **170**, the common electrode **445**, and the inner pad electrode **447** may be formed, and the plurality of second bonding electrodes **176** filling the plurality of via holes may be formed.

[0155] Referring to FIG. **19F**, after preparing the circuit board **420**, the circuit board **420** may be arranged on the resulting product of FIG. **19E** such that the second bonding insulating layer **162** and the plurality of second bonding electrodes **176** are arranged to face the first bonding insulating layer **440** and the plurality of first bonding electrodes **442** which are included in the circuit board **420**, and the circuit board **420** may be pressurized in an arrow AR direction on the surface exposing the second bonding insulating layer **162** and the plurality of second bonding electrodes **176** to bond the plurality of first bonding electrodes **442** to the plurality of second bonding electrodes **176** and to bond the first bonding insulating layer **440** to the second bonding insulating layer **162**.

[0156] The bonding between the plurality of first bonding electrodes **442** and the plurality of second bonding electrodes **176** and the bonding between the first bonding insulating layer **440** and the second bonding insulating layer **162** may be by wafer bonding, for example, the hybrid bonding described above.

[0157] Referring to FIG. **19G**, from the resulting product in which the plurality of first bonding electrodes **442** and the plurality of second bonding electrodes **176** are bonded and first bonding insulating layer **440** and the second bonding insulating layer **162** is bonded according to the process described in relation to FIG. **19F**, the substrate for growth **401** covering the first conductivity type base semiconductor layer **102** may be removed to expose the first conductivity type base semiconductor layer **102**. The substrate for growth **401** may be removed by various processes such as laser lift-off, mechanical polishing, mechanical-chemical polishing, or etching. After the first conductivity type base semiconductor layer **102** is exposed by removing the substrate for growth **401**, the thickness of the first conductivity type base semiconductor layer **102** may be reduced by using a polishing process such as chemical mechanical polishing (CMP), etc.

[0158] In the connection region CR and the contact pad region PAD, by partially etching the first conductivity type base semiconductor layer **102**, a part of the common electrode **445** and the inner pad electrode **447** may be exposed, and in the pixel region PXR, by partially etching the first conductivity type base semiconductor layer **102** and the reflective structure **150**, an electrode space in a grid shape or a mesh shape penetrating the first conductivity type base semiconductor layer **102** and the reflective structure **150** may be arranged. By filling a conductive material in the electrode space, the grid electrode **492** may be formed. In the pixel region PXR, the grid electrodes **492** may be formed to be connected to each other in a grid shape or a mesh shape.

[0159] The grid electrode **492** may be formed to include a part covering the rear surface **102B** of the first conductivity type base semiconductor layer **102** in the pixel region PXR and the connection region CR. The grid electrode **492** may be formed in contact with each of the first conductivity type base semiconductor layer **102** and the common electrode **445** in the connection region CR. A plating process may be used to form the grid electrode **492**; however, embodiments are not limited thereto.

[0160] As illustrated in FIG. **13**, the plurality of microlenses **496** covering the plurality of emission surfaces **102E** defined by the grid electrode **492** of the rear surface **102B** of the first conductivity type base semiconductor layer **102** may be formed.

[0161] In the contact pad region PAD, the contact pad electrode **494** may be formed on the inner pad electrode **447**, and by performing dicing in the edge region ISO (see FIG. **11**) of each of the plurality of adjacent modules, the display device **400** may be manufactured.

[0162] An example of a manufacturing method of the display device **400** illustrated in FIGS. **11**, **12**, and **13** is described in relation to FIGS. **19A** to **19G**; however, it may be obvious for a person skilled in the art that the light-emitting device **100**, **100A**, **100B**, **100C**, **100D**, **100E**, **100F**, and **200** illustrated in FIGS. **1** to **9B** may be manufactured by modifying or altering the description provided in relation to FIGS. **19A** to **19G** within the scope of the present disclosure.

[0163] FIG. 20 is a block diagram illustrating an example of an electronic device including a display device.

[0164] Referring to FIG. 18, an electronic device **8201** may be provided in a network environment **8200**. In the network environment **8200**, the electronic device **8201** may communicate with another electronic device **8202** through a first network **8298** (a short-range wireless communication network, etc.) or communicate with another electronic device **8204** and/or a server **8208** through a second network **8299** (a long-distance wireless communication network, etc.). The electronic device **8201** may communicate with the electronic device **8204** through the server **8208**. The electronic device **8201** may include a processor **8220**, a memory **8230**, an input device **8250**, an audio output device **8255**, a display device **8260**, an audio module **8270**, a sensor module **8276**, an interface **8277**, a haptic module **8279**, a camera module **8280**, a power management module **8288**, a battery **8289**, a communication module **8290**, a subscriber identification module **8296**, and/or an antenna module **8297**. In the electronic device **8201**, some of the constituent components may be omitted or other constituent components may be added. Some of these components may be implemented as one integrated circuit. For example, the sensor module **8276** (a fingerprint sensor, an iris sensor, an illuminance sensor, and the like) may be implemented by being embedded in the display device **8260** (a display, and the like).

[0165] The processor **8220** may control one or a plurality of other constituent elements (hardware and software constituent elements, and the like) of the electronic device **8201** connected to the processor **8220** by executing software (a program **8240**, and the like), and perform various data processing or calculations. As part of the data processing or calculations, the processor **8220** may load, in a volatile memory **8232**, commands and/or data received from other constituent elements (the sensor module **8276**, the communication module **8290**, and the like), process the command and/or data stored in the volatile memory **8232**, and store result data in a non-volatile memory **8234**. The processor **8220** may include a main processor **8221** (a central processing unit, an application processor, and the like) and an auxiliary processor **8223** (a graphics processing unit, an image signal processor, a sensor hub processor, a communication processor, and the like) that is operable independently of or together with the main processor **8221**. The auxiliary processor **8223** may use less power than the main processor **8221** and may perform a specialized function.

[0166] Instead of the main processor **8221** when the main processor **8221** is in an inactive state (sleep state), or with the main processor **8221** when the main processor **8221** is in an active state (application execution state), the auxiliary processor **8223** may control functions and/or states related to some constituent elements (the display device **8260**, the sensor module **8276**, the communication module **8290**, and the like) of the constituent elements of the electronic device **8201**. The auxiliary processor **8223** (an image signal processor, a communication processor, and the like) may be implemented as a part of functionally related other constituent elements (the camera module **8280**, the communication module **8290**, and the like).

[0167] The memory **8230** may store various data needed by the constituent elements (the processor **8220**, the sensor module **8276**, and the like) of the electronic device **8201**. The data may include, for example, software (the program **8240**, and the like) and input data and/or output data about commands related thereto. The memory **8230** may include the volatile memory **8232** and/or the non-volatile memory **8234**.

[0168] The program **8240** may be stored in the memory **8230** as software, and may include an operating system **8242**, middleware **8244**, and/or an application **8246**.

[0169] The input device **8250** may receive commands and/or data to be used for constituent elements (the processor **8220**, and the like) of the electronic device **8201**, from the outside (a user, and the like) of the electronic device **8201**. The input device **8250** may include a remote controller, a microphone, a mouse, a keyboard, and/or a digital pen (a stylus pen, and the like).

[0170] The audio output device **8255** may output an audio signal to the outside of the electronic device **8201**. The audio output device **8255** may include a speaker and/or a receiver. The speaker

may be used for general purposes such as multimedia playback or recording playback, and the receiver can be used to receive incoming calls. The receiver may be implemented by being coupled as a part of the speaker or by an independent separate device.

[0171] The display device **8260** may visually provide information to the outside of the electronic device **8201**. The display device **8260** may include a display, a hologram device, or a projector, and a control circuit to control a corresponding device. The display device **8260** may include the light-emitting device illustrated in FIG. **1** (**100**, **100A**, **100B**, **100C**, **100D**, **100E**, **100F**, or **200**) or the display device **400** illustrated in FIGS. **11**, **12**, and **13**. The display device **8260** may include a touch circuitry set to detect a touch and/or a sensor circuit (a pressure sensor, and the like) set to measure the strength of a force generated by the touch.

[0172] The audio module **8270** may convert sound into electrical signals or reversely electrical signals into sound. The audio module **8270** may obtain sound through the input device **8250**, or output sound through a speaker and/or a headphone of another electronic device (the electronic device **8202**, and the like) connected to the audio output device **8255** and/or the electronic device **8201** in a wired or wireless manner.

[0173] The sensor module **8276** may detect an operation state (power, temperature, and the like) of the electronic device **8201**, or an external environment state (a user state, and the like), and generate an electrical signal and/or a data value corresponding to a detected state. The sensor module **8276** may include a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an IR sensor, a biometric sensor, a temperature sensor, a humidity sensor, and/or an illuminance sensor.

[0174] The interface **8277** may support one or more specified protocols used for the electronic device **8201** to be connected to another electronic device (the electronic device **8202**, and the like) in a wired or wireless manner. The interface **8277** may include a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface.

[0175] A connection terminal **8278** may include a connector for the electronic device **8201** to be physically connected to another electronic device (the electronic device **8202**, and the like). The connection terminal **8278** may include an HDMI connector, a USB connector, an SD card connector, and/or an audio connector (a headphone connector, and the like).

[0176] The haptic module **8279** may convert electrical signals into mechanical stimuli (vibrations, movements, and the like) or electrical stimuli that are perceivable by a user through tactile or motor sensations. The haptic module **8279** may include a motor, a piezoelectric device, and/or an electrical stimulation device.

[0177] The camera module **8280** may capture a still image and a video. The camera module **8280** may include a lens assembly including one or more lenses, image sensors, image signal processors, and/or flashes. The lens assembly included in the camera module **8280** may collect light emitted from a subject for image capturing.

[0178] The power management module **8288** may manage power supplied to the electronic device **8201**. The power management module **8288** may be implemented as a part of a power management integrated circuit (PMIC).

[0179] The battery **8289** may supply power to the constituent elements of the electronic device **8201**. The battery **8289** may include non-rechargeable primary cells, rechargeable secondary cells, and/or fuel cells.

[0180] The communication module **8290** may establish a wired communication channel and/or a wireless communication channel between the electronic device **8201** and another electronic device (the electronic device **8202**, the electronic device **8204**, the server **8208**, and the like), and support a communication through an established communication channel. The communication module **8290** may be operated independent of the processor **8220** (the application processor, and the like), and may include one or more communication processors supporting a wired communication and/or a

wireless communication. The communication module **8290** may include a wireless communication module **8292** (a cellular communication module, a short-range wireless communication module, a global navigation satellite system (GNSS) communication module, and the like), and/or a wired communication module **8294** (a local area network (LAN) communication module, a power line communication module, and the like). Among the above communication modules, a corresponding communication module may communicate with another electronic device through the first network **8298** (a short-range communication network such as Bluetooth, WiFi Direct, or infrared data association (IrDA)) or the second network **8299** (a long-range communication network such as a cellular network, the Internet, or a computer network (LAN, WAN, and the like)). These various types of communication modules may be integrated into one constituent element (a single chip, and the like), or may be implemented as a plurality of separate constituent elements (multiple chips). The wireless communication module **8292** may verify and authenticate the electronic device **8201** in a communication network such as the first network **8298** and/or the second network **8299** by using subscriber information (an international mobile subscriber identifier (IMSI), and the like) stored in the subscriber identification module **8296**.

[0181] The antenna module **8297** may transmit signals and/or power to the outside (another electronic device, and the like) or receive signals and/or power from the outside. An antenna may include an emitter formed in a conductive pattern on a substrate (a printed circuit board (PCB), and the like). The antenna module **8297** may include one or a plurality of antennas. When the antenna module **8297** includes a plurality of antennas, the communication module **8290** may select, from among the antennas, an appropriate antenna for a communication method used in a communication network such as the first network **8298** and/or the second network **8299**. Signals and/or power may be transmitted or received between the communication module **8290** and another electronic device through the selected antenna. Other parts (an RFIC, and the like) than the antenna may be included as a part of the antenna module **8297**.

[0182] Some of the constituent components of the electronic device **8201** may be connected to each other through a communication method between peripheral devices (a bus, a general purpose input and output (GPIO), a serial peripheral interface (SPI), a mobile industry processor interface (MIPI), and the like) and may mutually exchange signals (commands, data, and the like).

[0183] The command or data may be transmitted or received between the electronic device **8201** and the external electronic device **8204** through the server **8208** connected to the second network **8299**. The electronic devices **8202** and **8204** may be of a type that is the same as or different from the electronic device **8201**. All or a part of operations executed in the electronic device **8201** may be executed in one or more electronic devices (**8202**, **8204**, and **8208**). For example, when the electronic device **8201** needs to perform a function or service, the electronic device **8201** may request one or more electronic devices to perform part of the whole of the function or service, instead of performing the function or service. The one or more electronic devices receiving the request may perform additional function or service related to the request, and transmit a result of the performance to the electronic device **8201**. To this end, cloud computing, distributed computing, and/or client-server computing technology may be used.

[0184] The electronic device **8201** may be applied to various devices. Various components of the electronic device **8201** may be modified according to a function of the device, and other components suitable for performing the function of the device may be further added. Hereinafter, application examples of the electronic device **8201** are described.

[0185] FIG. **21** is a diagram illustrating an example of a mobile device as an application example of the electronic device. A mobile device **9100** may include a display device **9110**. The display device **9110** may include the light-emitting device illustrated in FIG. **1** (**100**, **100A**, **100B**, **100C**, **100D**, **100E**, **100F**, or **200**) or the display device **400** illustrated in FIGS. **11**, **12**, and **13**. The display device **9110** may have a foldable structure, for example, a multi-foldable structure.

[0186] FIG. **22** is a diagram illustrating an example of a head-up display device for vehicles as an

application example of the electronic device. The head-up display device **9200** for vehicles may include a display **9210** provided in an area of a vehicle and an optical path change member **9220** configured to change an optical path such that a driver can see an image generated at the display **9210**. The display **9210** may include the light-emitting device illustrated in FIG. **1** (**100**, **100A**, **100B**, **100C**, **100D**, **100E**, **100F**, or **200**) or the display device **400** illustrated in FIGS. **11**, **12**, and **13**.

[0187] FIG. **23** is a diagram illustrating an example of augmented reality glasses or virtual reality glasses as an application example of the electronic device. Augmented reality glasses (or virtual reality glasses) **9300** may include a projection system **9310** configured to form an image and a guide element **9320** configured to guide the image from the projection system **9310** to user's eyes. The projection system **9310** may include the light-emitting device illustrated in FIG. **1** (**100**, **100A**, **100B**, **100C**, **100D**, **100E**, **100F**, or **200**) or the display device **400** illustrated in FIGS. **11**, **12**, and **13**.

[0188] FIG. **24** is a diagram illustrating an example of a large-scale signage as an application example of the electronic device. A signage **9400** may include the display described in relation to FIG. **12**. The signage **9400** may include the light-emitting device illustrated in FIG. **1** (**100**, **100A**, **100B**, **100C**, **100D**, **100E**, **100F**, or **200**) or the display device **400** illustrated in FIGS. **11**, **12**, and **13**. The signage **9400** may be used in outdoor advertising using a digital information display, and may control contents, etc. of an advertisement through a network. The signage **9400** may be implemented by, for example, the electronic device described with reference to FIG. **20**.

[0189] FIG. **25** is a diagram illustrating an example of a wearable display as an application example of the electronic device. A wearable display **9500** may include the light-emitting device illustrated in FIG. **1** (**100**, **100A**, **100B**, **100C**, **100D**, **100E**, **100F**, or **200**) or the display device **400** illustrated in FIGS. **11**, **12**, and **13**. The wearable display **9500** may be implemented through the electronic device described in relation to FIG. **20**.

[0190] The light-emitting device (**100**, **100A**, **100B**, **100C**, **100D**, **100E**, **100F**, or **200**) illustrated in FIGS. **1** to **9B** and the display device **400** illustrated in FIGS. **11**, **12**, and **13** may be applied to various products, such as a rollable television, a stretchable display, etc., in addition to the electronic devices described above.

[0191] While aspects of embodiments have been particularly shown and described, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. A light-emitting device comprising: a semiconductor light-emitting structure comprising a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer which are sequentially provided in a first direction; and an electrode layer covering the second conductivity type semiconductor layer and spaced apart from the active layer in the first direction with the second conductivity type semiconductor layer therebetween, wherein the semiconductor light-emitting structure is configured to emit light having a wavelength (λ) within a range from about 490 nm to about 700 nm, and wherein a minimum distance from the electrode layer to the active layer in the first direction is greater than about 0.05λ and less than about 0.24λ .
2. The light-emitting device of claim 1, wherein the second conductivity type semiconductor layer comprises a first surface in contact with the active layer and a second surface in contact with the electrode layer, and the second surface is opposite to the first surface in the first direction, and wherein a thickness of the second conductivity type semiconductor layer in the first direction is greater than about 0.05λ and less than about 0.24λ .
3. The light-emitting device of claim 1, wherein the electrode layer comprises: a transparent

electrode layer in contact with the second conductivity type semiconductor layer; and a reflective electrode layer in contact with the transparent electrode layer and spaced apart from the second conductivity type semiconductor layer in the first direction with the transparent electrode layer therebetween.

4. The light-emitting device of claim 1, wherein the active layer comprises a first surface in contact with the first conductivity type semiconductor layer and a second surface in contact with the second conductivity type semiconductor layer, and wherein a minimum distance from the first surface to the second surface in the first direction is less than 300 nm.

5. The light-emitting device of claim 1, wherein a thickness of the active layer in the first direction is less than 0.4λ .

6. The light-emitting device of claim 1, wherein the wavelength (λ) is within a range from about 610 nm to about 650 nm.

7. The light-emitting device of claim 1, wherein the wavelength (λ) is within a range from about 510 nm to about 550 nm.

8. The light-emitting device of claim 1, wherein the active layer comprises a multi-quantum well layer comprising a first surface in contact with the first conductivity type semiconductor layer and a second surface in contact with the second conductivity type semiconductor layer, and wherein a thickness of the multi-quantum well layer in the first direction is less than 300 nm.

9. The light-emitting device of claim 1, wherein the active layer comprises: a first multi-quantum well layer comprising a first surface in contact with the first conductivity type semiconductor layer; a second multi-quantum well layer comprising a second surface in contact with the second conductivity type semiconductor layer and spaced apart from the first multi-quantum well layer in the first direction; and a spacer barrier layer between the first multi-quantum well layer and the second multi-quantum well layer, and wherein in the first direction, a thickness of the spacer barrier layer is greater than each of a thickness of the first multi-quantum well layer and a thickness of the second multi-quantum well layer.

10. The light-emitting device of claim 9, wherein at least one constituent material of the spacer barrier layer is identical to at least one constituent material provided in the first multi-quantum well layer and the second multi-quantum well layer, and wherein in the first direction, each of the thickness of the first multi-quantum well layer and the thickness of the second multi-quantum well layer is less than 300 nm.

11. The light-emitting device of claim 1, wherein the active layer has a quantum well structure comprising a pair structure of a quantum well layer and a quantum barrier layer in 1 to 15 cycles, and wherein the pair structure in the quantum well structure is selected from InGaN/GaN, InGaN/InGaN, InGaN/AlGaN, and InGaN/InAlGaN.

12. The light-emitting device of claim 1, wherein the semiconductor light-emitting structure has a width less than 100 μm in a second direction perpendicular to the first direction.

13. The light-emitting device of claim 1, further comprising a reflective structure covering a side wall of each of the first conductivity type semiconductor layer, the active layer, and the second conductivity type semiconductor layer which are provided in the semiconductor light-emitting structure, wherein the reflective structure comprises a distributed Bragg reflector (DBR).

14. The light-emitting device of claim 1, further comprising a reflective structure covering a side wall of each of the first conductivity type semiconductor layer, the active layer, and the second conductivity type semiconductor layer which are provided in the semiconductor light-emitting structure, wherein the reflective structure comprises a metal.

15. The light-emitting device of claim 1, further comprising: a first conductivity type base semiconductor layer integrally connected to the first conductivity type semiconductor layer and spaced apart from the active layer in the first direction with the first conductivity type semiconductor layer therebetween; and a microlens spaced apart from the first conductivity type semiconductor layer in the first direction with the first conductivity type base semiconductor layer

therebetween and configured to collect and emit light from the semiconductor light-emitting structure.

16. A light-emitting device comprising: a first conductivity type base semiconductor layer; a semiconductor light-emitting structure comprising a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer which are sequentially provided on a main surface of the first conductivity type base semiconductor layer in a first direction perpendicular to the main surface; a first electrode surrounding a side wall of the first conductivity type base semiconductor layer; a transparent electrode layer covering the second conductivity type semiconductor layer and spaced apart from the active layer in the first direction with the second conductivity type semiconductor layer therebetween; a reflective structure covering a side wall of each of the semiconductor light-emitting structure and the transparent electrode layer; a second electrode in contact with the transparent electrode layer and spaced apart from the second conductivity type semiconductor layer in the first direction with the transparent electrode layer therebetween; and a microlens in contact with an emission surface comprising a part of a rear surface opposite to the main surface of the first conductivity type base semiconductor layer and configured to collect and emit light from the semiconductor light-emitting structure, wherein the semiconductor light-emitting structure is configured to emit light having a wavelength (λ) within a range from about 490 nm to about 700 nm, wherein the semiconductor light-emitting structure has a width of 100 μm or less in a second direction parallel with the main surface of the first conductivity type base semiconductor layer, wherein a minimum distance from the active layer to the transparent electrode layer in the first direction is greater than 0.05λ and less than 0.24λ , wherein the active layer comprises at least one multi-quantum well layer, and wherein a thickness of the at least one multi-quantum well layer in the first direction is less than 300 nm.

17. The light-emitting device of claim 16, wherein the first electrode comprises a first portion in contact with the side wall of the first conductivity type base semiconductor layer and a second portion in contact with the rear surface of the first conductivity type base semiconductor layer to define the emission surface of the first conductivity type base semiconductor layer, and wherein the microlens is in contact with the second portion of the first electrode.

18. The light-emitting device of claim 16, wherein the second conductivity type semiconductor layer comprises a first surface in contact with the active layer and a second surface in contact with the transparent electrode layer, and the second surface is opposite to the first surface in the first direction, and wherein a thickness of the second conductivity type semiconductor layer in the first direction is greater than 0.05λ and less than 0.24λ .

19. The light-emitting device of claim 16, wherein the active layer comprises a first surface in contact with the first conductivity type semiconductor layer and a second surface in contact with the second conductivity type semiconductor layer, and wherein a thickness of the at least one multi-quantum well layer in the first direction is less than 300 nm.

20. A display device comprising: a circuit board comprising a driver circuit; and a pixel array on the circuit board and comprising a plurality of pixels, wherein the pixel array comprises: a first conductivity type base semiconductor layer comprising a main surface and a rear surface which are opposite to each other; a plurality of semiconductor light-emitting structures on the main surface of the first conductivity type base semiconductor layer and spaced apart from each other in a horizontal direction parallel with the main surface, each of the plurality of semiconductor light-emitting structures comprising a first conductivity type semiconductor layer, an active layer, and a second conductivity type semiconductor layer which are sequentially provided in a first direction perpendicular to the main surface; a first electrode comprising a metal layer having a grid shape penetrating the first conductivity type base semiconductor layer in the first direction; a plurality of transparent electrode layers covering the second conductivity type semiconductor layer in each of the plurality of semiconductor light-emitting structures; a reflective structure covering a side wall of each of the plurality of semiconductor light-emitting structures and the plurality of transparent

electrode layers; a plurality of second electrodes penetrating the reflective structure in the first direction and in contact with one transparent electrode layer selected from the plurality of transparent electrode layers; and a plurality of microlenses on the rear surface of the first conductivity type base semiconductor layer and overlapping the plurality of semiconductor light-emitting structures in the first direction, wherein at least one semiconductor light-emitting structure selected from the plurality of semiconductor light-emitting structures is configured to emit light having a wavelength (λ) within a range from about 490 nm to about 700 nm, and wherein in the at least one semiconductor light-emitting structure, a minimum distance from the active layer to a transparent electrode layer, from among the plurality of transparent electrode layers, in the first direction is greater than about 0.05λ and less than about 0.24λ .
